Recovering a Permutation Key with a Linear FIR Equalizer and Decoding a Scrambled Color Image

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***Abstract*—A binary permutation key that encrypts a color image is reconstructed by estimating a linear FIR equalizer from a 32-symbol preamble and then hard-detecting the key bits. The equalizer order is motivated with numerical and visual criteria. The image is decoded using the provided decoder algorithm. Robustness is studied by injecting random bit flips into the reconstructed key.**

***Index Terms*—Linear equalization, FIR filter, permutation coding, robustness.**

I. Assignment and Data

A scrambled RGB image *cPic*, a noisy sequence $r(k)$ carrying the key, and a known training preamble of length 32 with alphabet $\{-1,+1\}$ are provided. The tasks are:

(i) reconstruct the key using a suitable linear FIR equalizer and justify the filter order,
(ii) decode the image with the recovered key,
(iii) determine how many bit errors the key can contain before decoding becomes visually impossible.

II. Solution Method

A. Signal model and detector

Bits are mapped to antipodal symbols $b(k)\in\{-1,+1\}$. The received sequence is modeled as a noisy finite-memory convolution:

$$r(k)=\sum_{l=0}^{L_h} h(l)\,b(k-l)+n(k), \qquad n(k)\sim\mathcal{N}(0,\sigma^2).$$

A causal FIR equalizer of order L forms

$$z(k)=\sum_{\ell=0}^{L} w(\ell)\,r(k-\ell), \qquad \hat{b}(k)=\operatorname{sign}\big(z(k)\big). \tag{1}$$

B. Training window and matrix formulation

The first 32 symbols are known training bits $b(k)$. For a fixed order L, the training indices are $k=L{+}1,\dots,32$. Which yields $N_{\mathrm{tr}}=32-L$ equations. Define the tap vector

$$\mathbf{w}=[w(0),w(1),\dots,w(L)]^{\mathsf{T}}\in\mathbb{R}^{L+1},$$

the target vector

$$\mathbf{b}=\big[\,b(L{+}1),b(L{+}2),\dots,b(32)\,\big]^{\mathsf{T}}\in\{-1,+1\}^{N_{\mathrm{tr}}},$$

and the regression matrix

$$\mathbf{R}=\begin{bmatrix} r(L{+}1) & r(L) & \cdots & r(1)\\ r(L{+}2) & r(L{+}1) & \cdots & r(2)\\ \vdots & \vdots & & \vdots\\ r(32) & r(31) & \cdots & r(32{-}L)\end{bmatrix}\in\mathbb{R}^{N_{\mathrm{tr}}\times(L+1)}.$$

On the training window, (1) becomes $\mathbf{z}=\mathbf{R}\mathbf{w}$ with $\mathbf{z}\in\mathbb{R}^{N_{\mathrm{tr}}}$.

C. Least-squares estimation

The equalizer is estimated by

$$\hat{\mathbf{w}}=\arg\min_{\mathbf{w}}\big\|\mathbf{R}\mathbf{w}-\mathbf{b}\big\|_2^2. \tag{2}$$

If $\mathbf{R}^{\mathsf{T}}\mathbf{R}$ is invertible (overdetermined case), the solution is

$$\hat{\mathbf{w}}=(\mathbf{R}^{\mathsf{T}}\mathbf{R})^{-1}\mathbf{R}^{\mathsf{T}}\mathbf{b}. \tag{3}$$

In general, the minimum–norm solution is $\hat{\mathbf{w}}=\mathbf{R}^{+}\mathbf{b}$ with the Moore–Penrose pseudoinverse $\mathbf{R}^{+}$. With $\hat{\mathbf{w}}$ fixed, detection on the full record uses $\hat{b}(k)=\operatorname{sign}\big(\sum_{\ell=0}^{L}\hat{w}(\ell)\,r(k-\ell)\big)$. The first 32 decisions are then replaced by the known preamble.

D. Filter-order selection and quality measures

The sweep is conducted over $L=0,1,\dots,15$, which ensures an overdetermined training system because $32-L\ge L+1$. For each order the following quantities are computed on the training indices $\mathcal{K}_{\mathrm{tr}}=\{L{+}1,\dots,32\}$:

$$E_{\mathrm{tr}}(L)\triangleq\sum_{k\in\mathcal{K}_{\mathrm{tr}}}\mathbf{1}\big\{\operatorname{sign}\big(z(k)\big)\neq b(k)\big\}\quad\text{(training errors)}, \tag{4}$$

$$\mathrm{MSE}_{\mathrm{tr}}(L)\triangleq\frac{1}{|\mathcal{K}_{\mathrm{tr}}|}\sum_{k\in\mathcal{K}_{\mathrm{tr}}}\big(z(k)-b(k)\big)^2\quad\text{(training MSE)}. \tag{5}$$

To verify that the detected bits induce an almost one-to-one permutation, the decoder's bit-to-decimal mapping is replicated. Post-preamble bits are reshaped into a matrix of $(x{+}y{+}z)$ rows and p=10 columns. Each row is interpreted as an MSB→LSB binary word and converted to a decimal. The first x decimals correspond to row indices, the next y to column indices, and the last z to color channels. Uniqueness counts $U_x(L),U_y(L),U_z(L)$ are the numbers of distinct decimals in the three segments, and

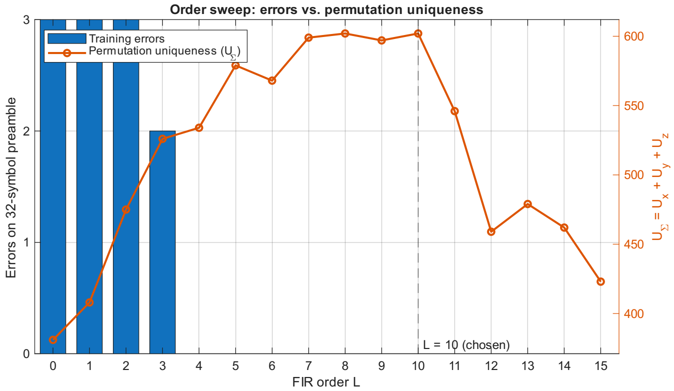

Fig. 1: Order sweep over $L = 0, \ldots, 15$: training errors $E_{\rm tr}(L)$ (bars, left axis) and permutation uniqueness $U_\Sigma(L)$ (line, right axis).

TABLE I: Selection metrics at the chosen order $L = 10$.

Training errors $E_{\rm tr}(10)$	0
Training MSE $\mathrm{MSE}_{\rm tr}(10)$	5.21×10^{-2}
$U_x(10)$ (target $= x$)	331/350
$U_y(10)$ (target $= y$)	268/277
$U_z(10)$ (target $= z$)	3/3

the aggregate score is $U_\Sigma(L) = U_x(L) + U_y(L) + U_z(L)$. Orders are ranked by:

$$\min E_{\rm tr}(L) \;\rightarrow\; \max U_\Sigma(L) \;\rightarrow\; \min \mathrm{MSE}_{\rm tr}(L).$$

Figure 1 compares $E_{\rm tr}(L)$ and $U_\Sigma(L)$ across orders. The best trade-off occurs at $L = 10$. The numerical values at the selected order are summarized in Table I.

III. Decoding

Using the selected order L=10, the full sequence is equalized and hard decisions are taken. The first 32 bits are replaced by the known preamble, and the resulting key feeds the provided decoder. To justify the order choice visually, Figure 2 compares decoded images for representative FIR orders.

IV. Robustness

Random errors are injected *after* the preamble. Let $N_{\rm key} = (x+y+z)P$ denote the payload key length in bits (x=350, y=277, z=3, precision P=10 $\Rightarrow N_{\rm key}$=6300). For a given number of flipped bits f, the relative corruption is the bit error rate

$$\mathrm{BER} = \frac{f}{N_{\rm key}}.$$

Figure 3 shows one realization per level with $f \in \{0, 10, 50, 200, 500, 1000\}$, i.e., BER $\in$ $\{0,\ 0.16\%,\ 0.79\%,\ 3.17\%,\ 7.94\%,\ 15.87\%\}$. Up to about 3% BER the content remains interpretable; between 3% and 8% the quality degrades rapidly. Beyond roughly 8% the image becomes visually unusable.

V. Conclusions

A 32-symbol preamble is sufficient to recover the permutation key by least-squares training of a causal FIR equalizer. An order sweep with a transparent selection rule—minimize preamble errors, then maximize permutation–uniqueness, then minimize MSE—selects L=10. With this order the preamble is detected without errors. The key exhibits near-full uniqueness in x and y (full in z), and the decoder produces a clear image. Robustness tests indicate that the picture remains interpretable up to roughly 3% BER and becomes unusable beyond approximately 8% BER. The assignment objectives—justified order selection, successful decoding, and quantified error tolerance—are satisfied with concise numerical and visual evidence.

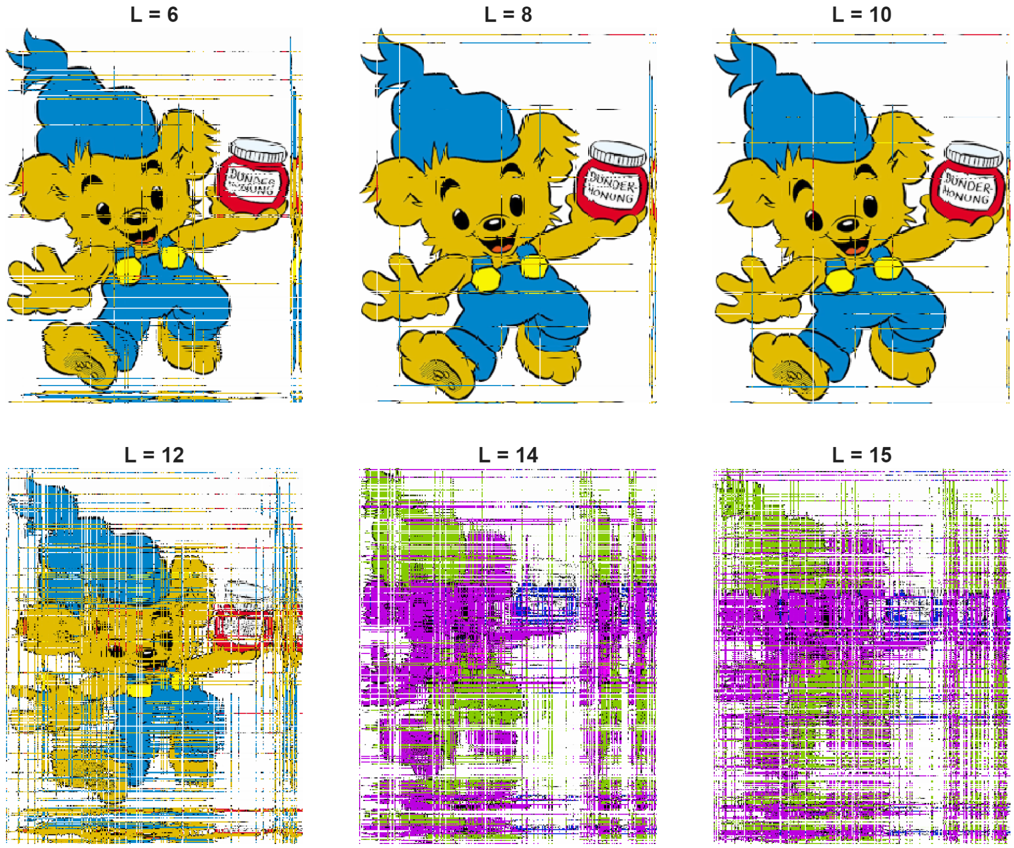

Fig. 2: Decoding with different equalizer orders.

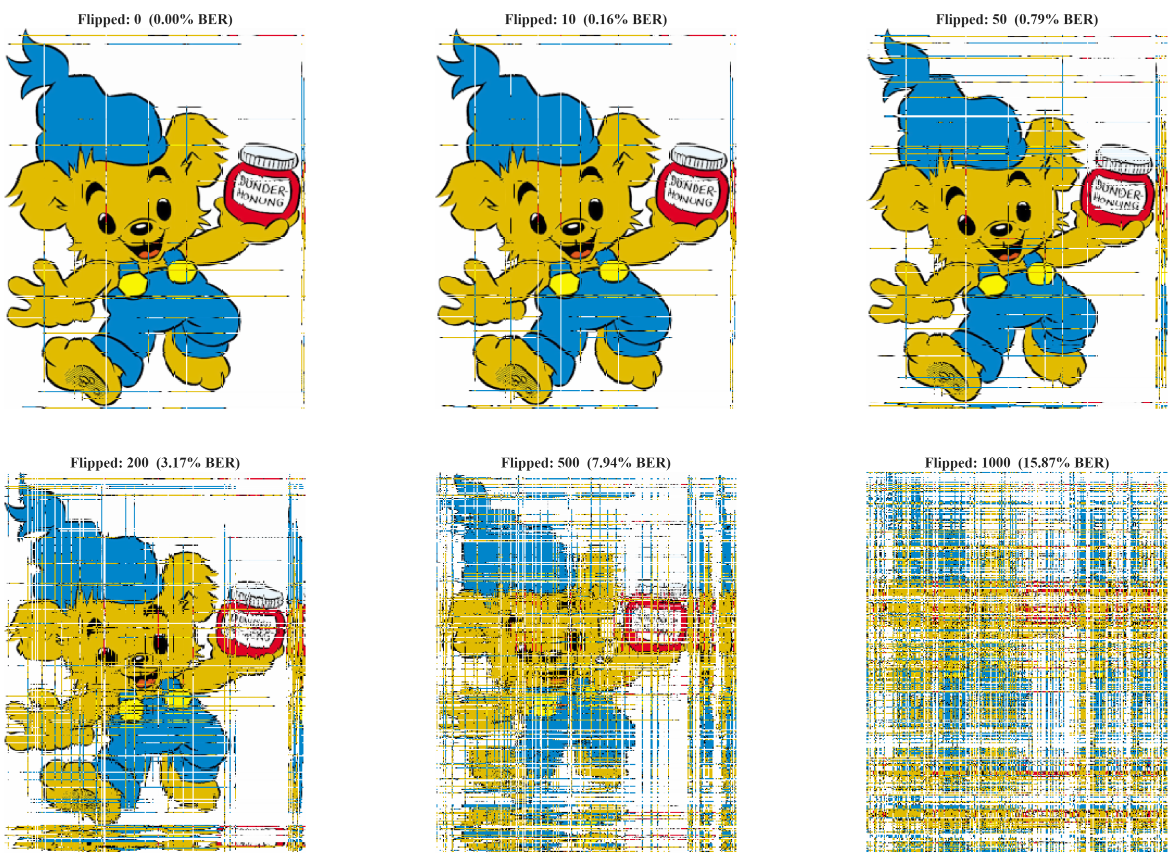

Fig. 3: Robustness to key errors.